Modification of pressure perturbation correction

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Purpose

The numerical scheme is modified in order to overcome the unphysical time step limitations $\delta t_{\max} = \sigma_{AN} \frac{(BR)^2}{n_e(T_e + T_i)}$ associated with

- Semi-implicit numerical scheme;
- Presence of artificial current $j_{AN} = -\sigma_{AN} \nabla \varphi$

Description of the instability

The characteristic times mentioned above appear due to interplay of pressure perturbation redistribution by E×B drift and radial diamagnetic current redistribution inside the separatrix. Since the artificial anomalous current may give the main contribution to the radial current on closed flux surfaces, it defines the value of radial electric field and the corresponding poloidal drift velocity. Thus the characteristic time depends on the artificial conductivity:

$$\tau = \sigma_{AN} \frac{(BR)^2}{n_e(T_e + T_i)}$$

The estimate for maximal value of the time step then comes as simple Courant-Friedrichs-Lewi criterion:

$$\delta t_{\max} = \sigma_{AN} \frac{(BR)^2}{n_e(T_e + T_i)}$$

The numerical instability manifests itself in simultaneous oscillations at the time step of pressure perturbation and of radial electric field on the closed flux surfaces.

Solution

It is suggested to reduce the poloidal pressure perturbation by an artificial enhancement of the poloidal spreading of particles and energy keeping the particle and energy balance. This is controlled by the following newly introduced parameters $\alpha_a < 1$, $\alpha_T < 1$, $\alpha_T \sim \alpha_a$, as described below.

The application of this correction to numerical scheme would allow to increase the time step by the factor of $\alpha_T^{-1} \sim \alpha_{a\,main}^{-1}$.

How does it work?

First, the corrections for plasma component densities $\delta n_{al,m}^{k}$ (*a* – sort of ions, *k* – time step index, *l* - radial and *m* - poloidal indexes), electron and ion temperatures $\delta T_{el,m}^{k}$, $\delta T_{il,m}^{k}$ are calculated routinely at the time step. Next, inside the separatrix the average corrections:

$$\left\langle \delta n_{al}^{k} \right\rangle = \frac{\sum_m \delta n_{al,m}^{k} \sqrt{g_{l,m}}}{\sum_m \sqrt{g_{l,m}}}$$

$$\left\langle \delta T_{el}^{k} \right\rangle = \frac{\sum_m \delta T_{el,m}^{k} n_{el,m} \sqrt{g_{l,m}}}{\sum_m n_{el,m} \sqrt{g_{l,m}}}$$

$$\left\langle \delta T_{il}^{k} \right\rangle = \frac{\sum_{a,m} \delta T_{il,m}^{k} n_{al,m} \sqrt{g_{l,m}}}{\sum_{a,m} n_{al,m} \sqrt{g_{l,m}}}$$

are calculated on each flux surface and applied to each cell. Such averaging gives the same increase/decrease of the full number of particles and stored energy in electron and ion components of plasma in the cells at chosen flux surface as initial correction. Then, the corrections for densities and temperatures perturbations are calculated and applied with a scaling coefficients $\alpha_a < 1$, $\alpha_T < 1$, $\alpha_T \sim \alpha_a$. At the beginning of the next time-step the densities and temperatures are:

$$n_{al,m}^{k+1} = n_{al,m}^{k} + \left\langle \delta n_{al}^{k} \right\rangle + \alpha_a (\delta n_{al,m}^{k} - \left\langle \delta n_{al}^{k} \right\rangle)$$

$$T_{el,m}^{k+1} = T_{el,m}^{k} + \left\langle \delta T_{el}^{k} \right\rangle + \alpha_T (\delta T_{el,m}^{k} - \left\langle \delta T_{el}^{k} \right\rangle)$$

$$T_{il,m}^{k+1} = T_{il,m}^{k} + \left\langle \delta T_{il}^{k} \right\rangle + \alpha_T (\delta T_{il,m}^{k} - \left\langle \delta T_{il}^{k} \right\rangle)$$

After the application of modified correction the density is checked to ensure that it does not fall below the limiting value $n_{a\min}$ which is determined through the **b2mn.dat** as *b2mndr_na_min*. In case if it is not determined explicitly, $n_{a\min} = 10^4 m^{-3}$. If some of the density values for species *a* at the flux surface are below the limit after application of modified correction, the modification of correction cancels out and the correction is applied without modification for species *a* at this flux surface. This check ensures that values of density for the small impurity fractions, which can have big density variation at the flux surface and small time scales, stay positive after the application of correction.

The modification of correction is applied only inside the separatrix in the volume cells (not applied in the narrow boundary cells at the core boundary). It damps the effect of the redistribution of particles at the flux surface due to the E×B drift and permits the increase of time-step roughly by $\alpha_T^{-1} \sim \alpha_{a\,main}^{-1}$ times. In case $\alpha_a = \alpha_T = 1$ the modification of correction disappears.

This scheme does not affect the converged solution. In principle, in case of the solution with small density perturbation on closed flux surfaces (no cold X-point), the evolution of plasma can be also traced with this scheme applied. The evolution of pedestal and transport barrier in this case is determined mainly by radial diffusion and heat transport processes, while the time-scale of perturbation adjustment is anyway much smaller with or without modification of corrections. The coefficients α_a for small fractions of impurity (the criteria is $n_a(Z_a T_e + T_i) << n_{amain}(T_e + T_i)$) can be chosen equal to unity, since the time scales for pressure redistribution at the flux surface for small amount of impurity are big and the pressure perturbation influences neoclassical radial transport. The instability depends on full pressure and will not be affected by these impurities. The scheme should be treated with accuracy for the modeling of the discharges in a state of deep detachment, where the density and pressure perturbation at the flux surfaces next to X-point can be of the order of unity.

The boundary cells should be excluded from the correction due to following reasons. The usual boundary condition is (in its numerical realization) of type I. For example for density it is realized through the solution of equation:

$$n_{al,m}^{k+1} = n_{al,m}^{k} - \beta(n_{al,m}^{k+1} - n_0)$$

where n_0 is prescribed value and β is big coefficient.

If no correction is applied then $n_{al,m}^{k+1} = \dfrac{n_{al,m}^{k} + \beta n_0}{1+\beta} \xrightarrow[\beta\to\infty]{} n_0$, and that is necessary result.

Modified subroutines

The modification of the numerical scheme is applied in the subroutines **b2upco.F** and **b2upht.F**, also affected are **b2npc9.F** and **b2npco.F**.

The coefficients α_a, α_T are defined in **b2mod_numerics_namelist.F**.

Restrictions and application limits

- The application of new numerical scheme converges to the same steady state solution as one can get without this scheme, i.e. after reaching the converged steady state solution no additional run with modified input parameters is needed.
- The new numerical scheme has been successfully tested on cases with one main (hydrogenic) species and fluid neutrals (no impurities).
- The new numerical scheme has been successfully tested on cases with one main (hydrogenic) species plus impurities with kinetic neutrals.
- The scheme should be applied with care if the strong poloidal density and/or temperature variation are expected in the solution, e.g. for detached plasma with cold X-point.

Practical recommendations

In `b2mn.dat` increase the time step

```
'#b2mndr_dtim'      '1.0e-7'
'b2mndr_dtim'      '1.0e-5'
```

In `b2mn.dat` set up the desired minimal density value

```
'b2mndr_na_min'                    '1.0e8'
```

In `b2.numerics.parameters` set up the control parameters for each species and for temperature (example is given for `D+N` plasma)

```
corr_core_dn(0)=10*0.01,
corr_core_dt=0.01,
```

Here stands **`corr_core_dn`** for α_a and **`corr_core_dt`** stands for α_T.